

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

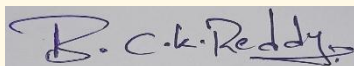
ASSESSMENT TOOL 2 – Case Study

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	CSE – Ai / 2024	Date:	10/08/2026

Code of Conduct

I B. Charan Kumar Reddy (192472195) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

11. Edge Detection in Low Contrast Images

An image has low contrast, causing edge detection algorithms to fail.

Analyze the problem and explain how preprocessing enhances edge detection results.

Case Analysis

An image with low contrast has only a small difference between the intensity values of neighboring pixels. Edge detection algorithms such as **Sobel and Canny** depend on intensity changes or gradients to identify object boundaries. When the contrast is low, these intensity changes become weak, making true edges difficult to distinguish from the background and noise.

Therefore, directly applying an edge detector to a low-contrast image may result in **missing edges, weak edges, broken boundaries, or false edges**.

1. Understanding of the Problem

The main problem is that the image does not have sufficient intensity variation between objects and their background.

For example:

- A dark-gray object on a slightly lighter-gray background may have a very weak boundary.
- In a medical image, tissues with similar intensities may have boundaries that are difficult to detect.

The edge detector requires sufficient intensity difference to produce a strong gradient. Therefore, **contrast enhancement should be performed before edge detection**.

2. Application of Image Processing Concepts

The following preprocessing techniques can improve edge detection:

A. Contrast Enhancement

Contrast enhancement increases the difference between dark and bright regions.

CLAHE (Contrast Limited Adaptive Histogram Equalization) is particularly suitable because it improves local contrast while limiting excessive noise amplification.

B. Noise Reduction

A small amount of noise reduction can be applied using a **Gaussian filter** before edge detection.

This reduces unwanted noise that could otherwise be detected as false edges.

C. Edge Detection

After preprocessing, the **Canny edge detector** can be applied to identify strong and continuous object boundaries.

3. Proposed Solution

The recommended processing pipeline is:

Low-Contrast Image

↓

Grayscale Conversion

↓

CLAHE Contrast Enhancement

↓

Gaussian Blur

↓

Canny Edge Detection

↓

Enhanced Edge Image

Why this sequence?

First, CLAHE improves the local contrast. Gaussian filtering then reduces noise introduced or emphasized during enhancement. Finally, Canny detects the stronger intensity transitions produced by the preprocessing.

4. Analysis

Without preprocessing:

Low Contrast → Weak Intensity Difference → Weak Gradients → Missing/Incomplete Edges

With preprocessing:

CLAHE → Improved Local Contrast → Stronger Intensity Difference → Better Gradients → Clearer Edges

Thus, preprocessing improves the ability of the edge detector to distinguish object boundaries from the background.

However, excessive contrast enhancement should be avoided because it can amplify noise and create unwanted edges. Similarly, excessive Gaussian smoothing can remove fine details. Therefore, the parameters must be selected carefully.

5. Recommended Technique: CLAHE + Canny

CLAHE

CLAHE divides the image into small local regions and enhances the contrast within each region. It also limits excessive histogram amplification, making it more suitable for low-contrast images than simple global enhancement in many cases.

Canny Edge Detection

Canny detects edges using image gradients and applies thresholding and edge tracking to obtain clear boundaries.

Therefore:

CLAHE → improves the visibility of weak boundaries

Gaussian Filter → reduces noise

Canny → detects the enhanced boundaries

6. Justification of the Solution

The proposed **CLAHE + Gaussian filtering + Canny** approach is suitable because:

1. **CLAHE improves local contrast**, making weak edges more visible.
2. **Gaussian filtering reduces noise**, preventing unnecessary false edges.
3. **Canny provides reliable edge detection** after the contrast has been improved.
4. The method is computationally practical and can be implemented using OpenCV.
5. The approach can be applied to medical images, surveillance images, industrial inspection, and other low-contrast computer vision applications.

7. Expected Result

Before Preprocessing

The original low-contrast image is expected to produce:

- Weak edges
- Missing boundaries
- Broken edges

- Possible false detections

After Preprocessing

The enhanced image is expected to produce:

- Stronger edges
- More continuous boundaries
- Better separation between objects and background
- Improved edge detection accuracy

8. Conclusion

Low contrast makes edge detection difficult because neighboring pixels have only small intensity differences. Direct application of an edge detector may therefore fail to identify important boundaries.

The proposed solution is to perform **CLAHE-based contrast enhancement followed by Gaussian noise reduction and Canny edge detection**. This improves local contrast, strengthens useful intensity transitions, reduces noise, and produces clearer object boundaries.

Therefore, preprocessing is an important step for improving edge detection performance in low-contrast images.

Final Processing Pipeline

Low-Contrast Image → CLAHE → Gaussian Blur → Canny Edge Detection → Clearer Edges